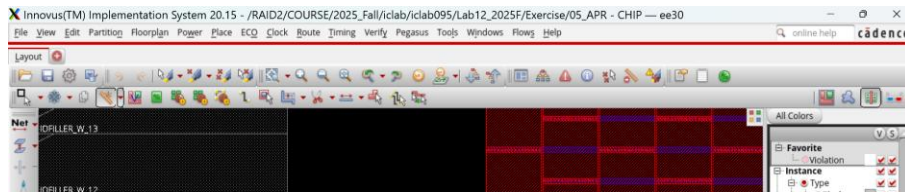


Report

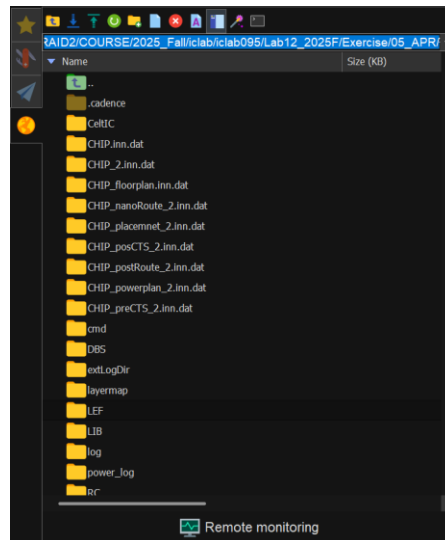
1. server info



2. server info

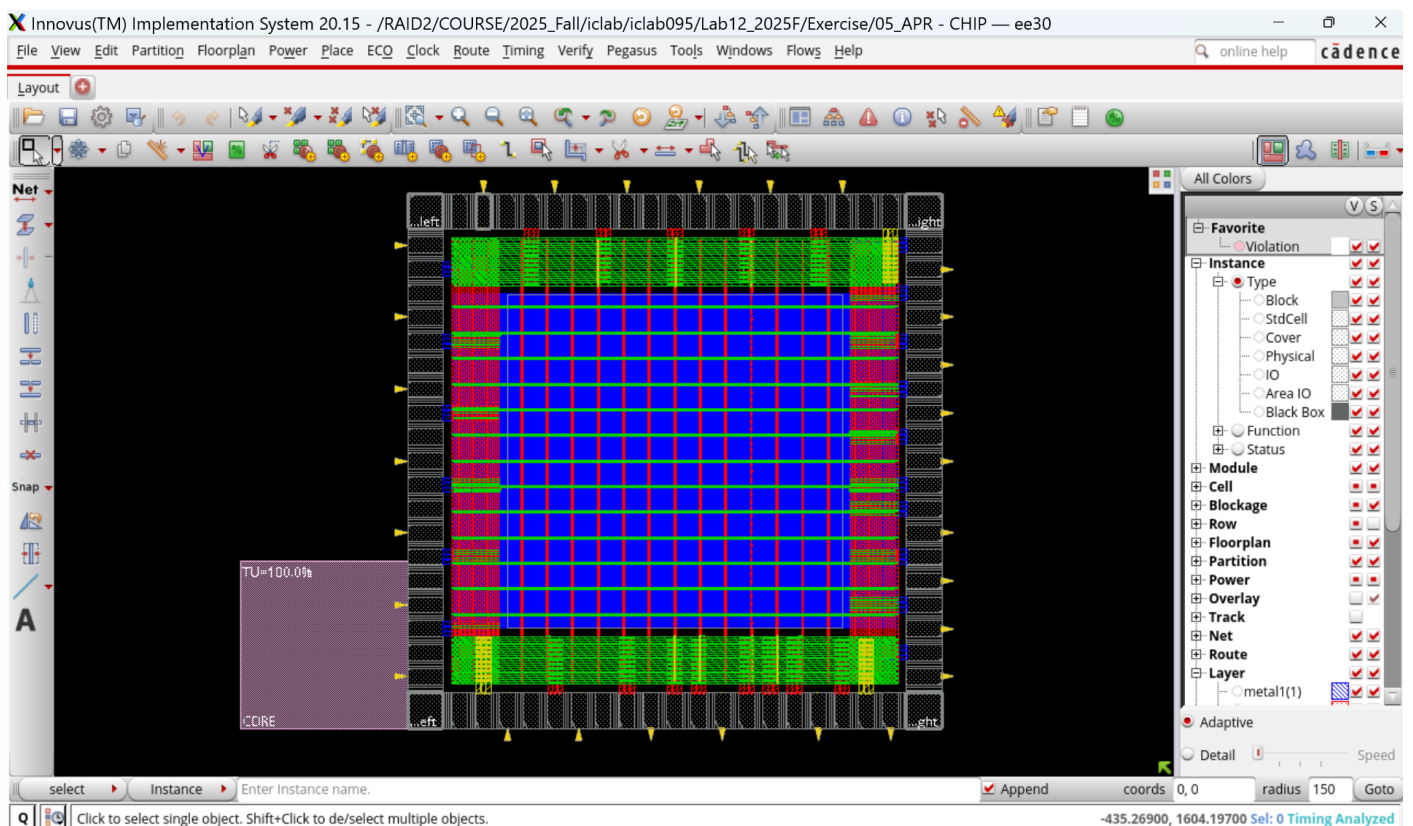
```
21:16 iclab095@ee30[~/Lab12_2025F/Exercise]$ cd 05_APR/  
21:16 iclab095@ee30[~/Lab12_2025F/Exercise/05_APR]$ setenv OA_HOME /RAID2/cad/cadence/INNOVUS  
/INNOVUS_20.15.000/share/oa  
21:17 iclab095@ee30[~/Lab12_2025F/Exercise/05_APR]$ innovus
```

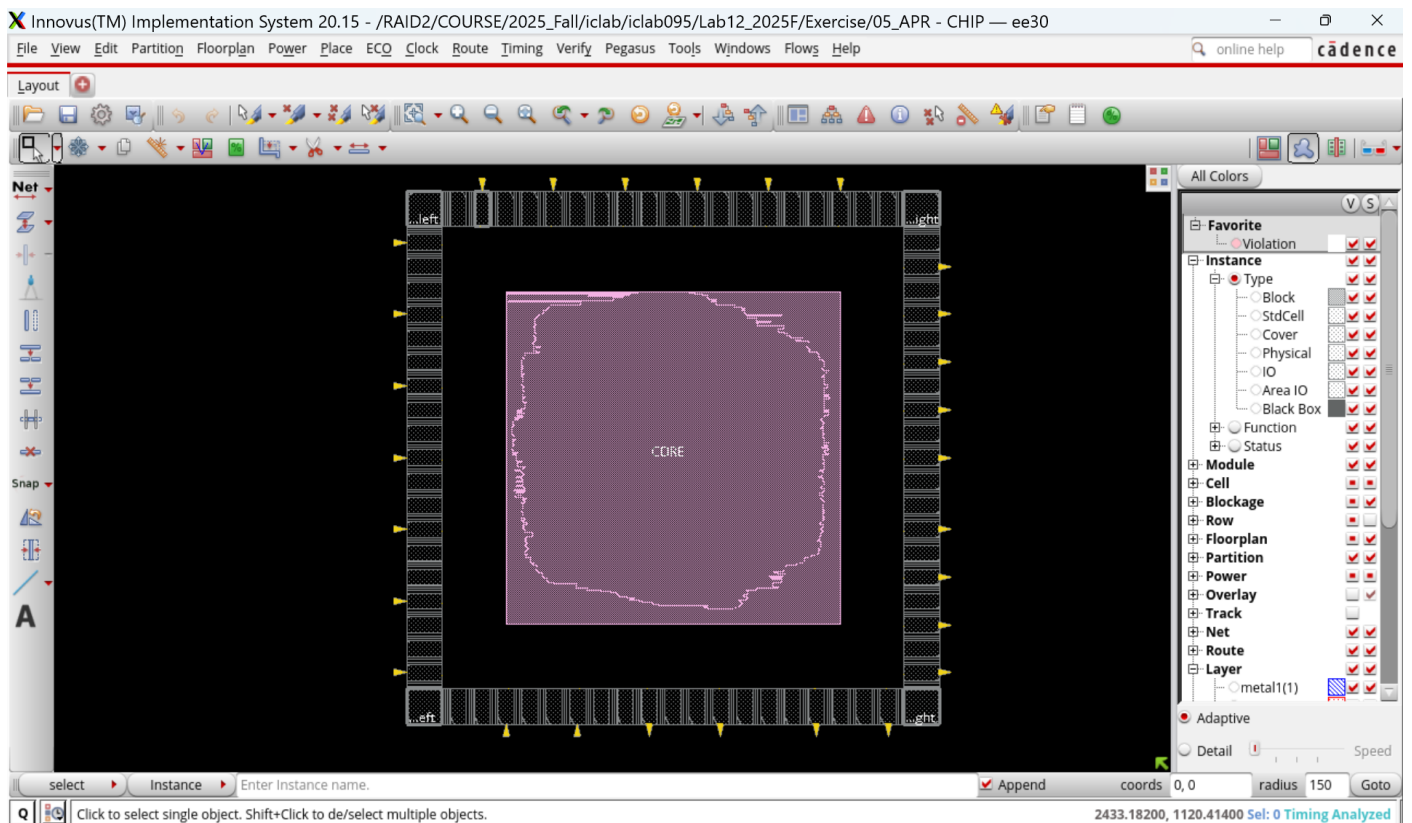
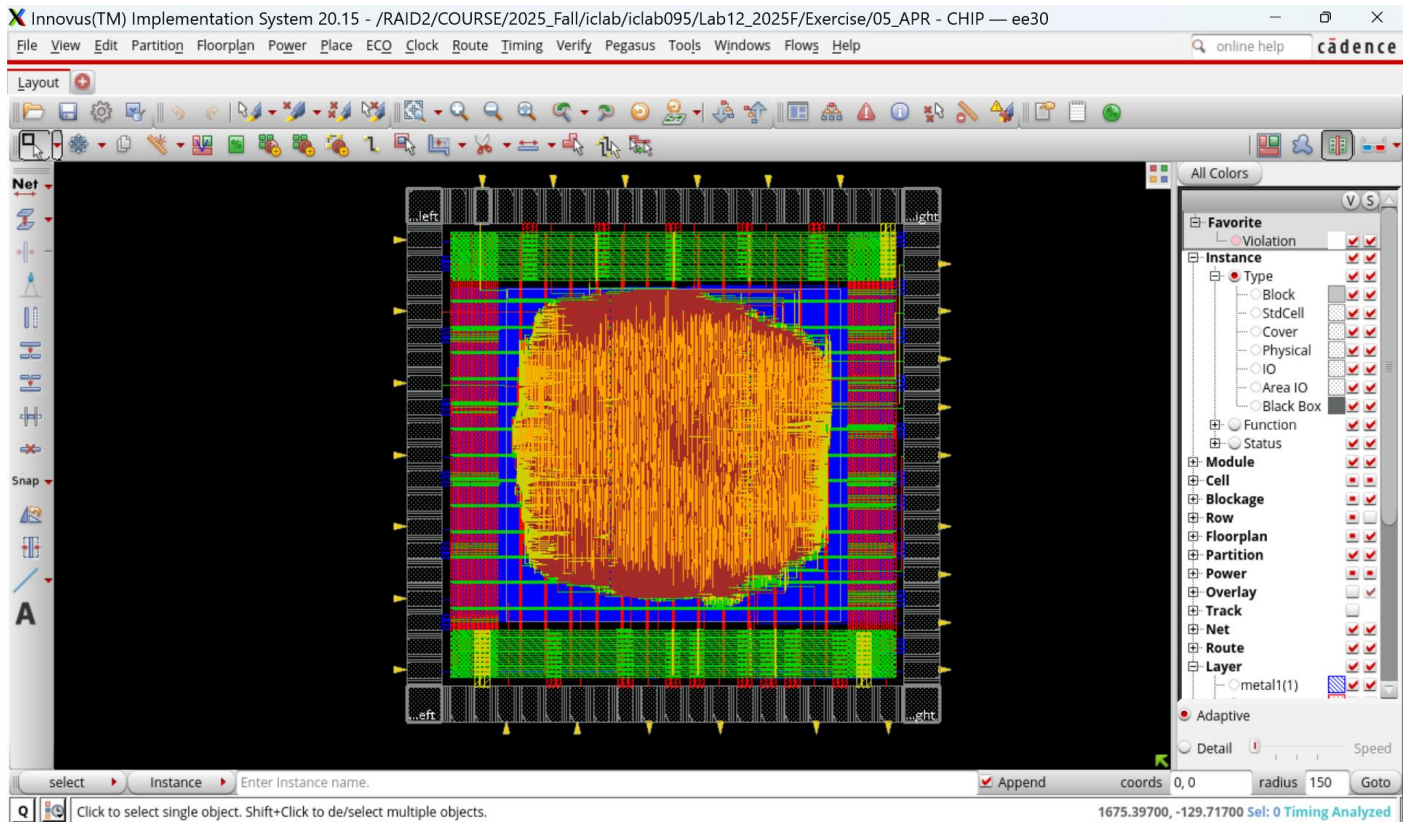
3. server info



1. Chip Layout View

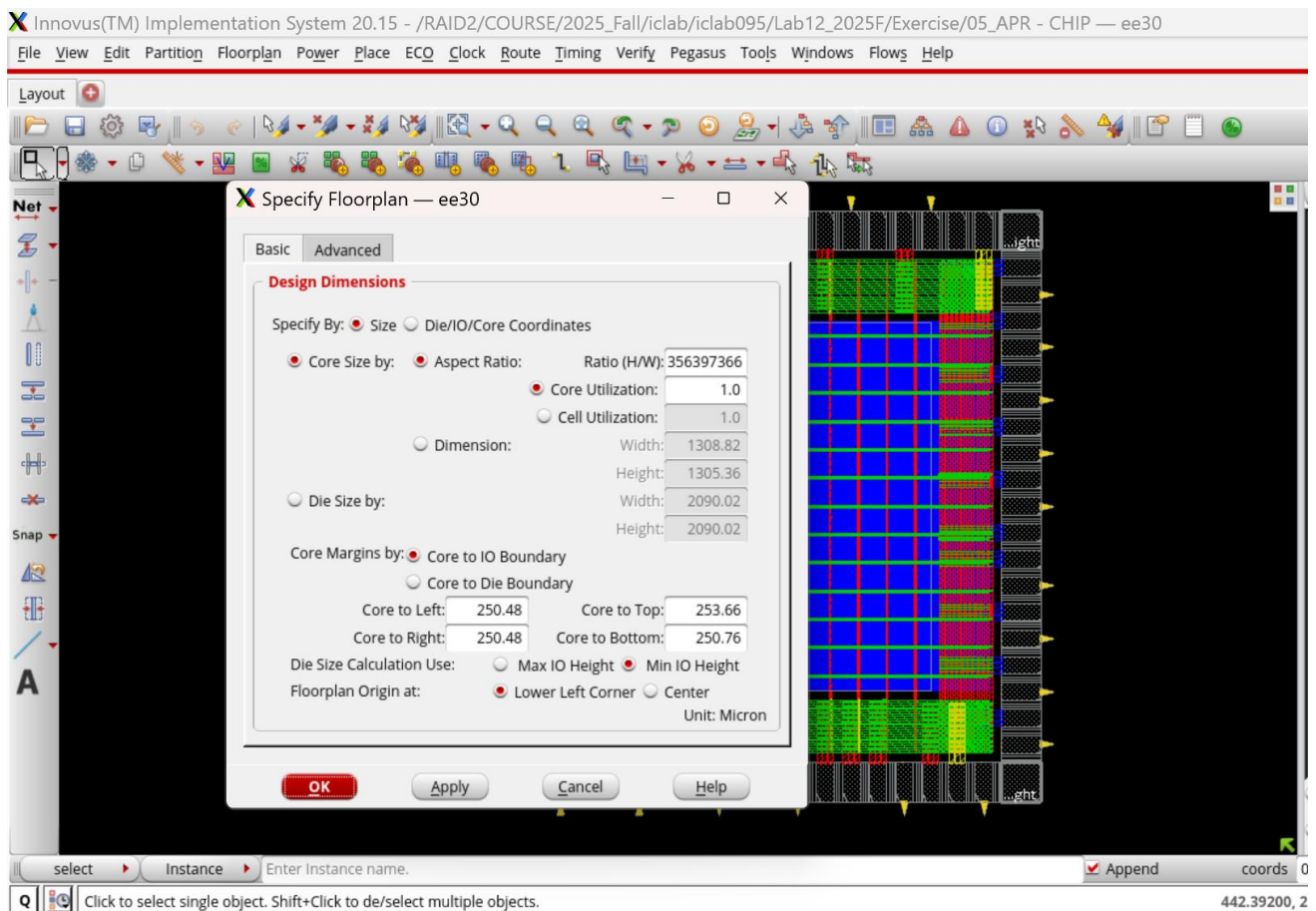
Instructions: After completing the final layout, capture a screenshot of the entire chip layout.





2. Core to IO boundary

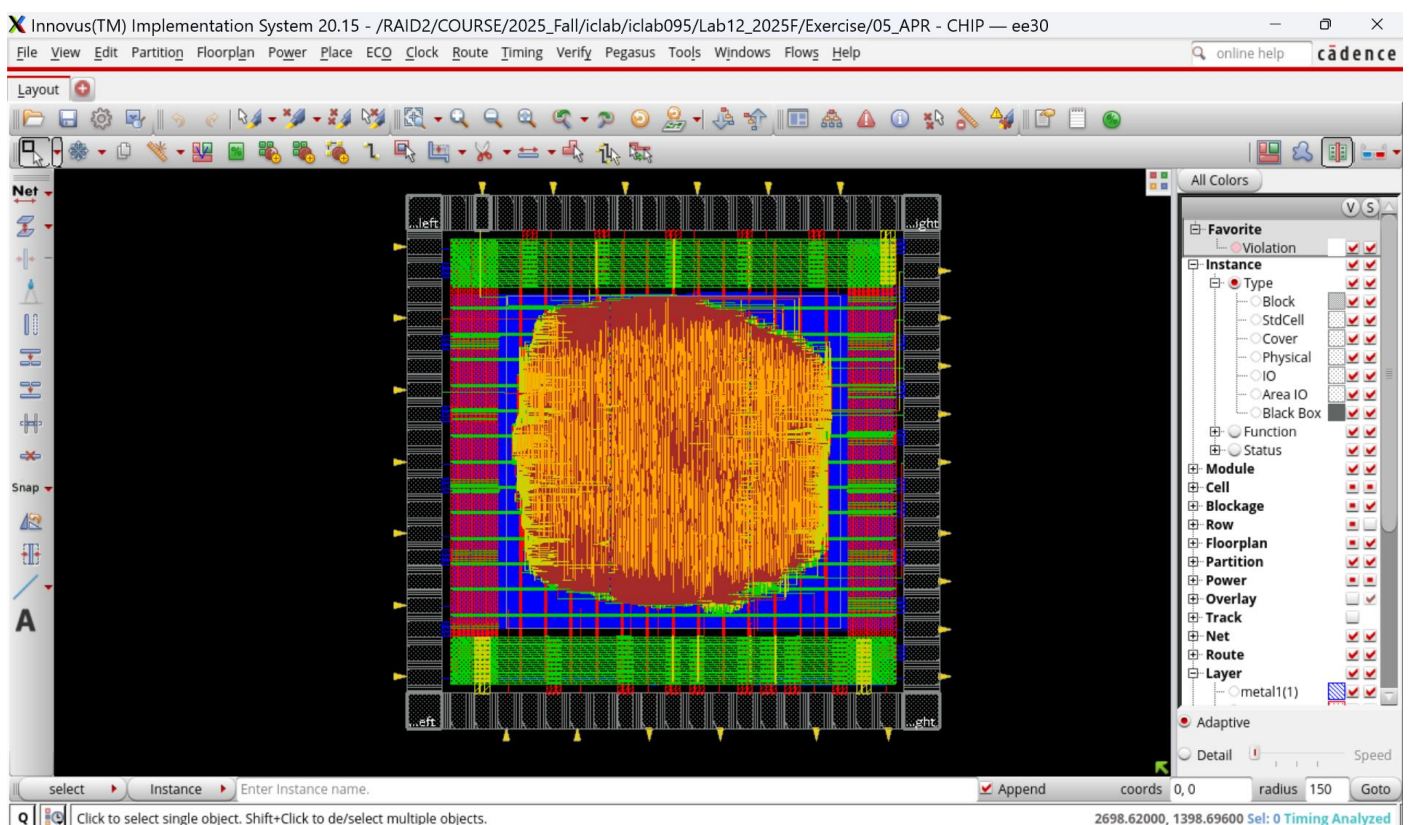
Instructions: After completing the final layout, open Floorplan -> Specify Floorplan, and capture a screenshot.



3. Core Ring

Instructions: After completing the final layout, zoom in on the Power section (ensuring the entire power ring is visible), and use the K key to measure the width of a single power line.

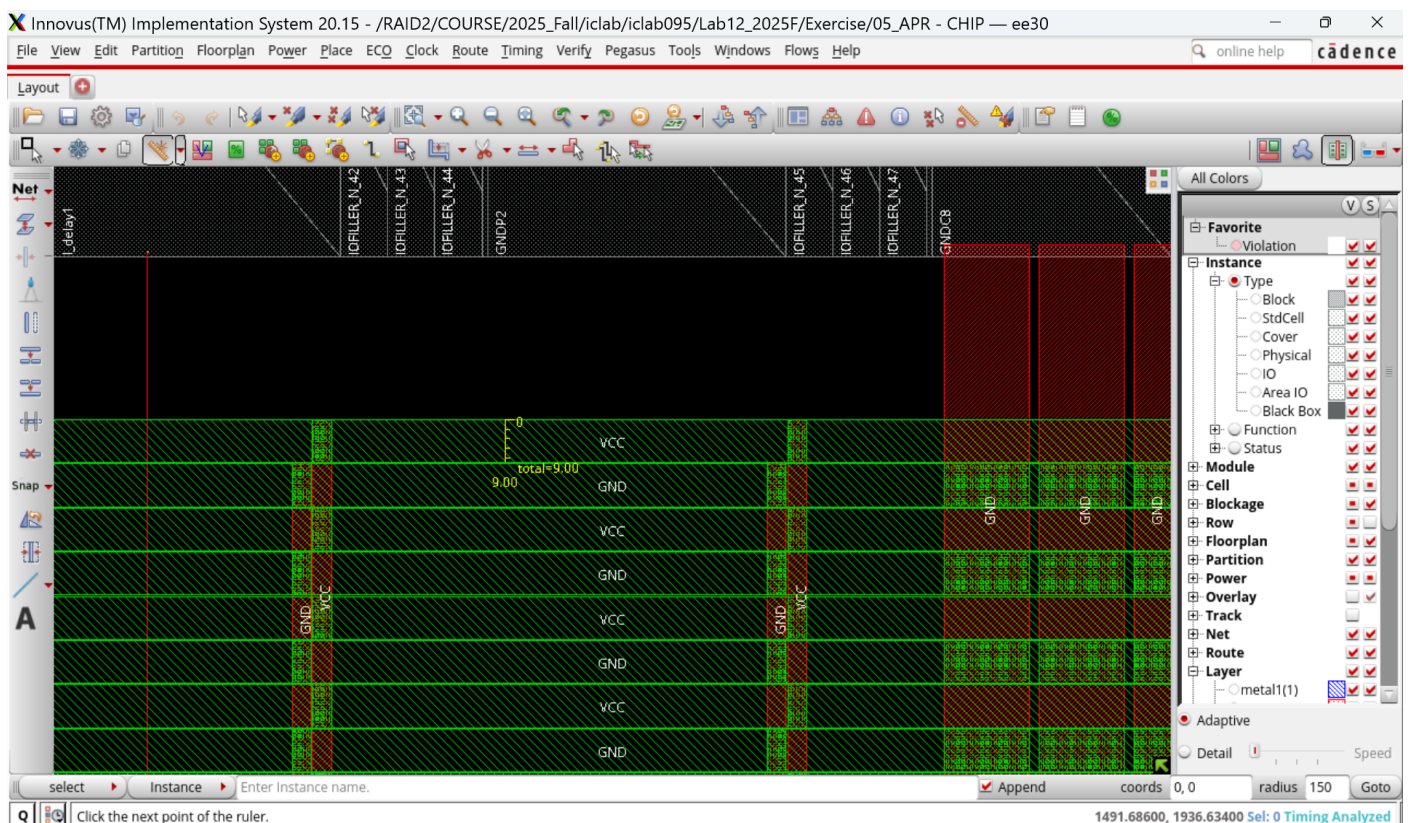
Overall Power Ring



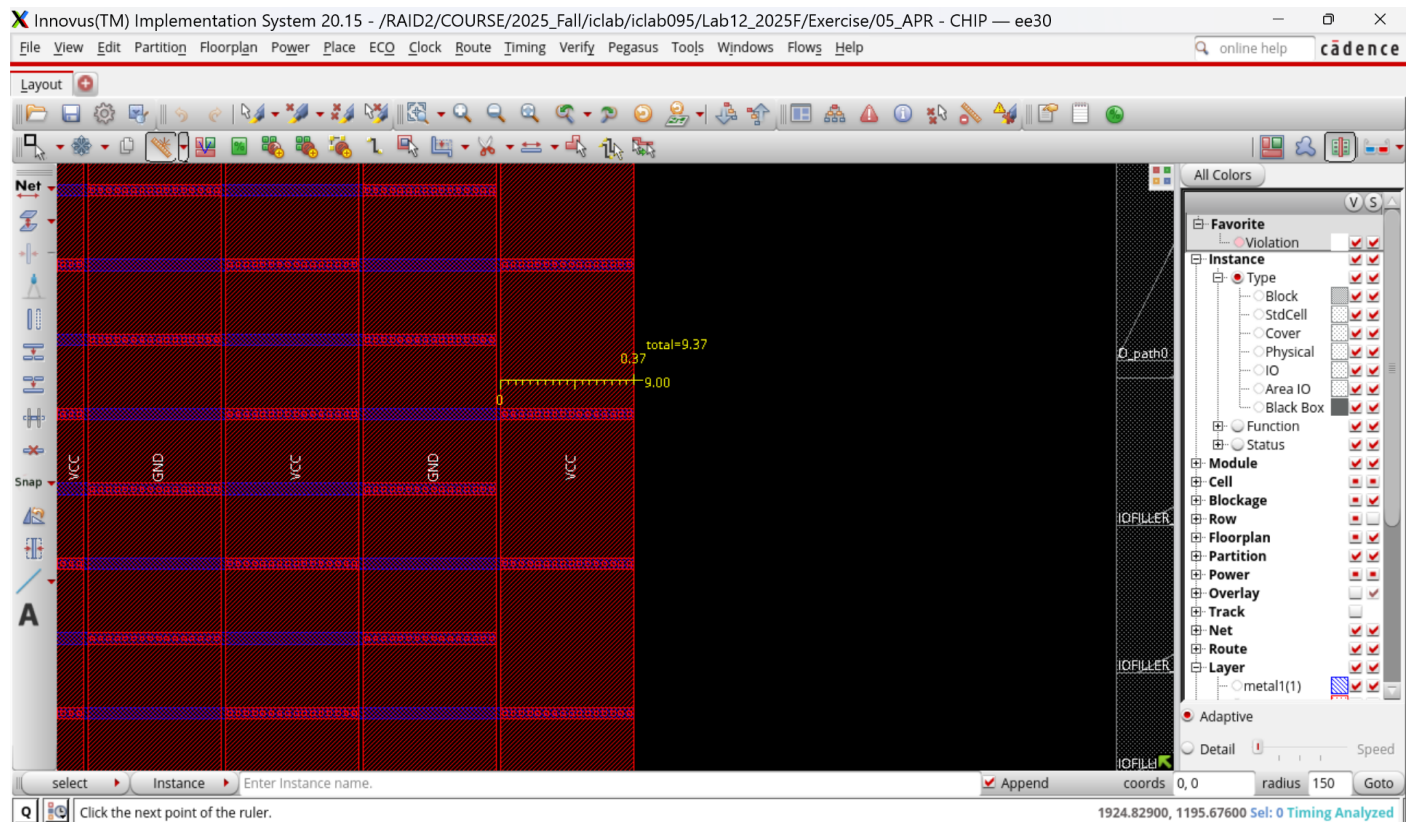
Overall Power Ring: wire group, interleaving, 10 pairs, width 9.



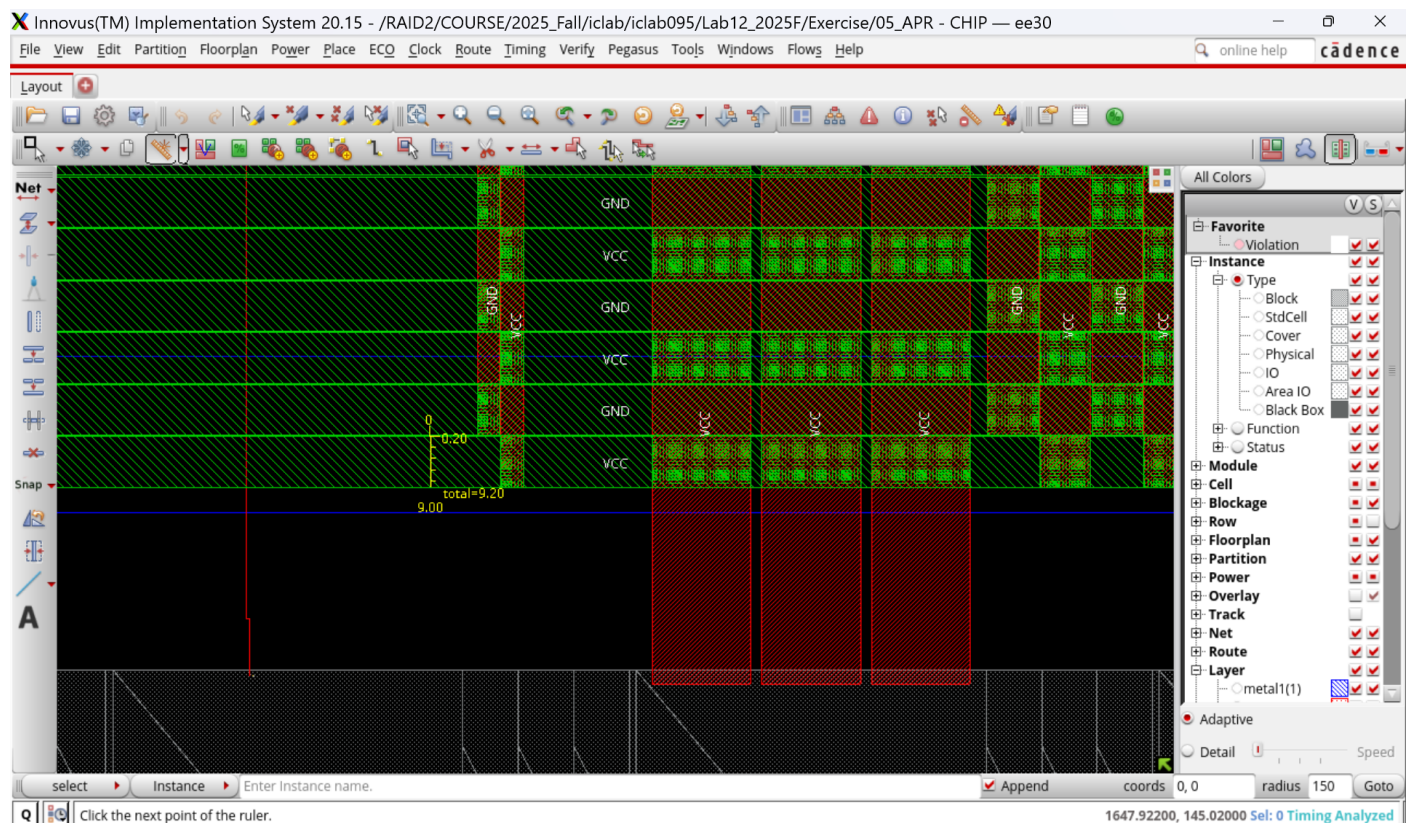
Top Power Ring: width=9



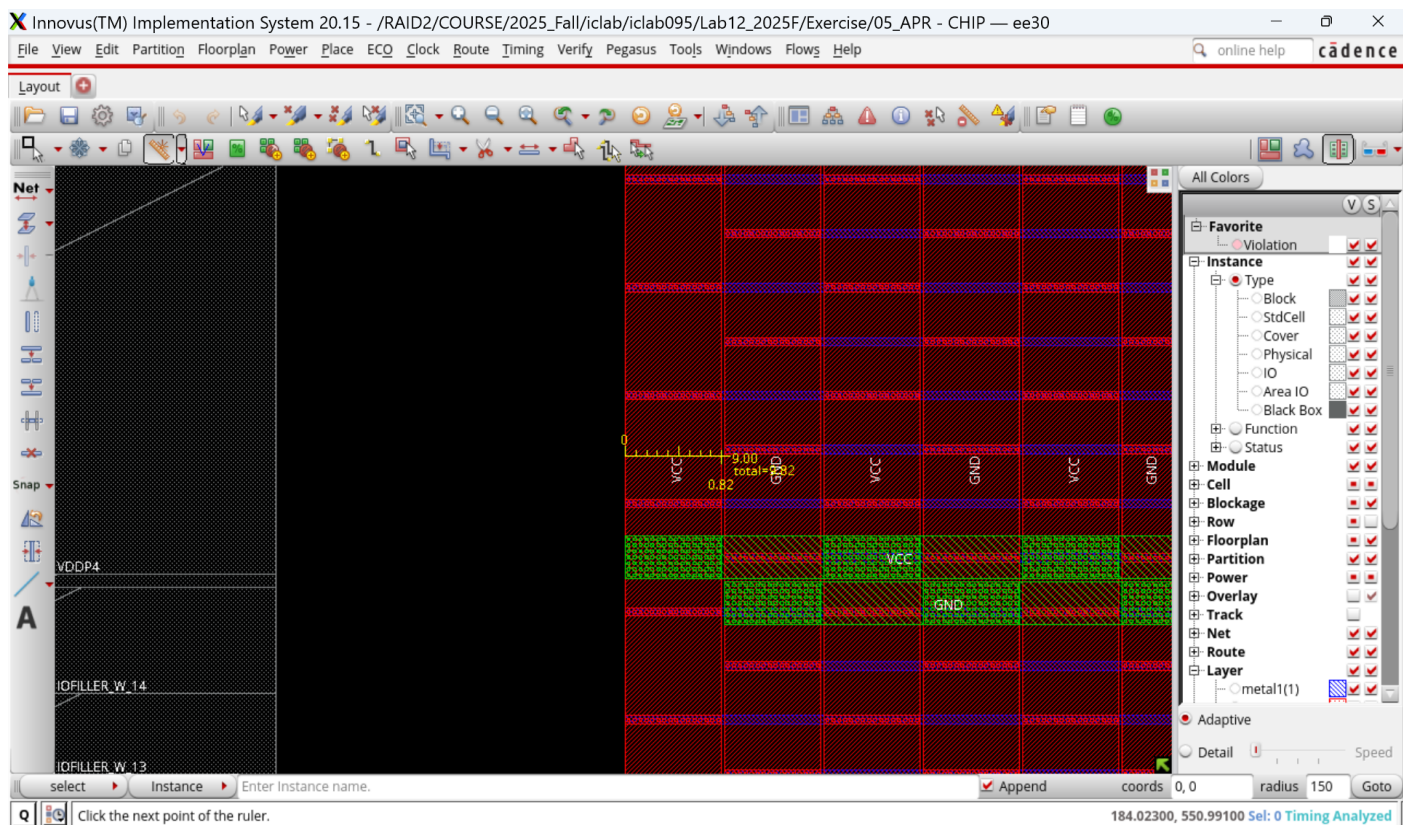
Right Power Ring: width=9



Bottom Power Ring: width=9



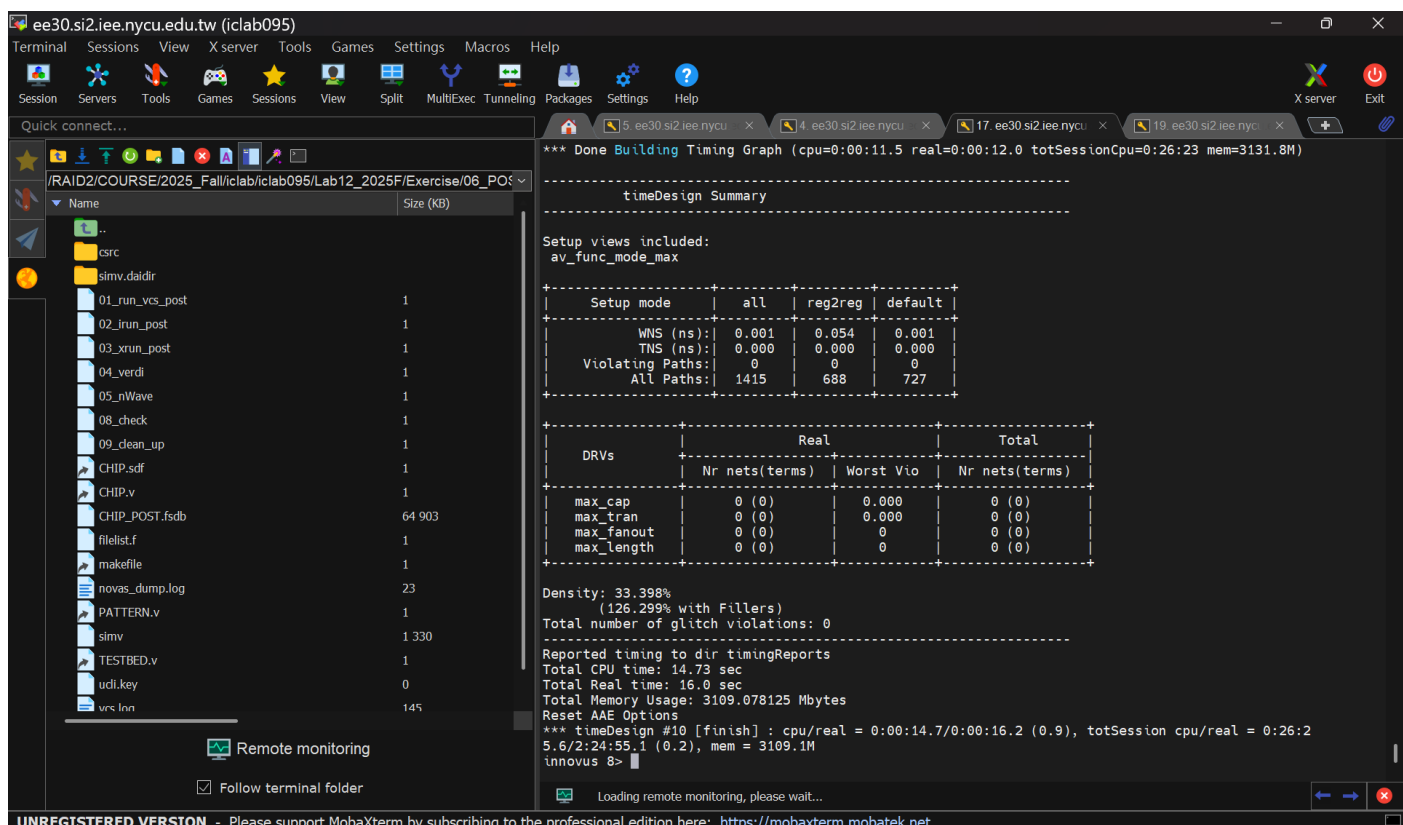
Left Power Ring: width=9



(Width settings are all 9)

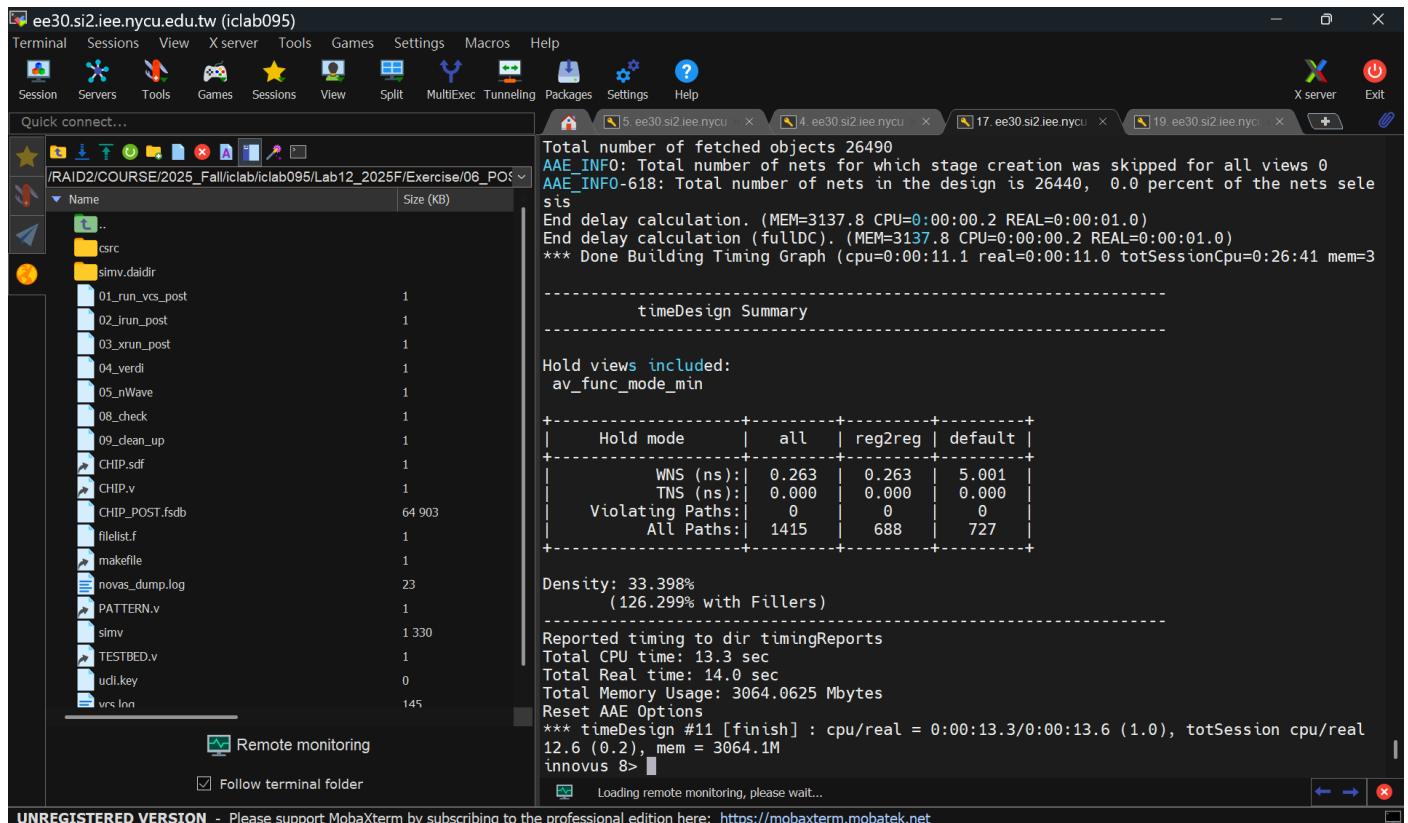
4. Post-Route setup time analysis :

Instructions: After completing the analysis, capture a screenshot of the results displayed in the terminal (including the timeDesign Summary).



5. Post-Route hold time analysis

Instructions: After completing the analysis, capture a screenshot of the results displayed in the terminal (including the timeDesign Summary).



The screenshot shows a MobaXterm window with a terminal session. The left pane displays a file explorer view of the directory /RAID2/COURSE/2025_Fall/iclab/iclab095/Lab12_2025F/Exercise/06_PO. The right pane shows the output of a timing analysis command. The output includes a 'timeDesign Summary' section with a table of hold mode results.

```
Total number of fetched objects 26490
AAE_INFO: Total number of nets for which stage creation was skipped for all views 0
AAE_INFO-618: Total number of nets in the design is 26440, 0.0 percent of the nets sele
sis
End delay calculation. (MEM=3137.8 CPU=0:00:00.2 REAL=0:00:01.0)
End delay calculation (fullDC). (MEM=3137.8 CPU=0:00:00.2 REAL=0:00:01.0)
*** Done Building Timing Graph (cpu=0:00:11.1 real=0:00:11.0 totSessionCpu=0:26:41 mem=3

-----
timeDesign Summary
-----

Hold views included:
av_func_mode_min

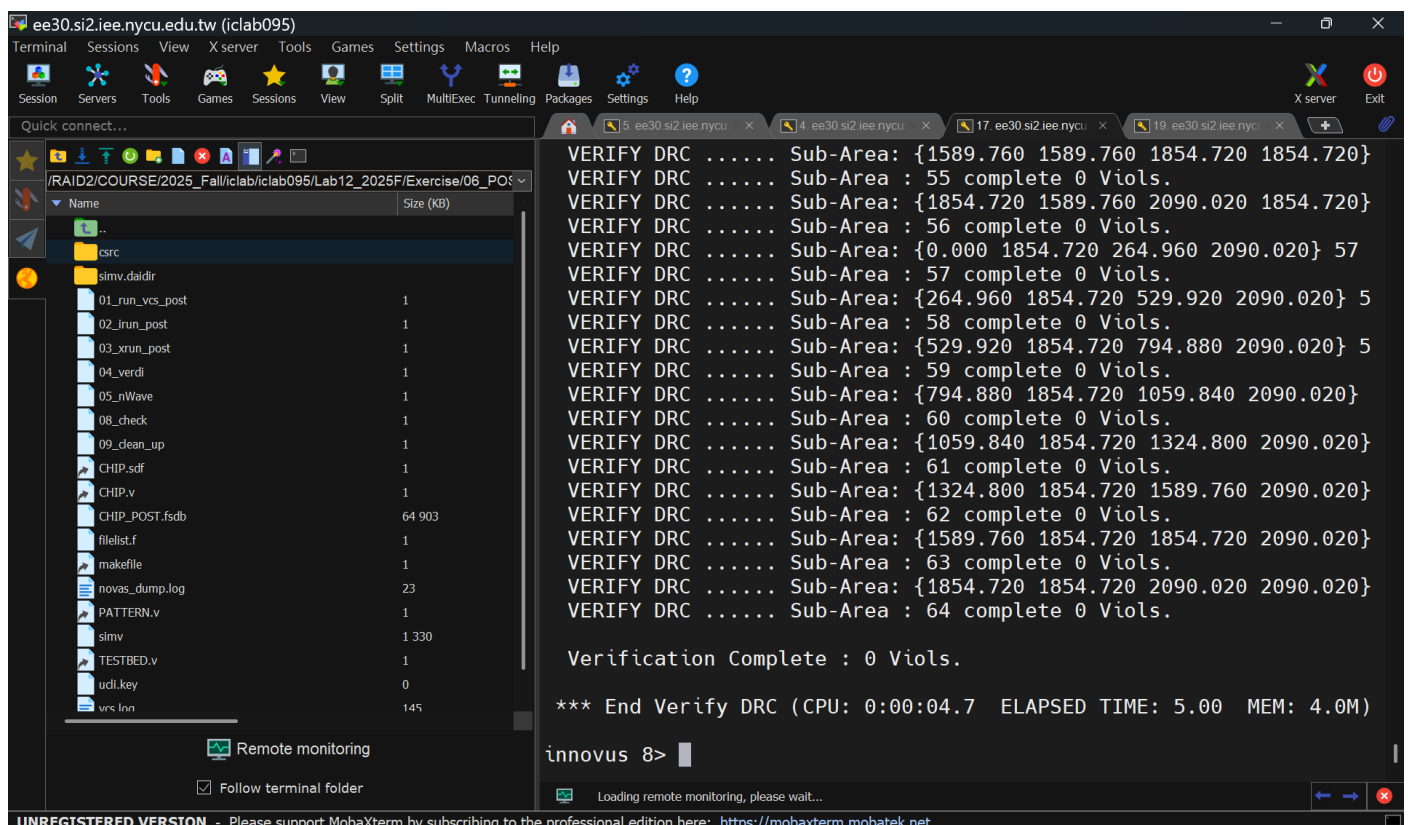
+-----+-----+-----+-----+
| Hold mode | all | reg2reg | default |
+-----+-----+-----+-----+
| WNS (ns): | 0.263 | 0.263 | 5.001 |
| TNS (ns): | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 0 | 0 | 0 |
| All Paths: | 1415 | 688 | 727 |
+-----+-----+-----+-----+

Density: 33.398%
(126.299% with Fillers)

-----
Reported timing to dir timingReports
Total CPU time: 13.3 sec
Total Real time: 14.0 sec
Total Memory Usage: 3064.0625 Mbytes
Reset AAE Options
*** timeDesign #11 [finish] : cpu/real = 0:00:13.3/0:00:13.6 (1.0), totSession cpu/real
12.6 (0.2), mem = 3064.1M
innovus 8>
```

6. DRC result

Instructions: After completing the final layout, run the analysis and capture a screenshot of the results displayed in the terminal (including the number of violations).



The screenshot shows a MobaXterm window with a terminal session. The left pane displays a file explorer view of the directory /RAID2/COURSE/2025_Fall/iclab/iclab095/Lab12_2025F/Exercise/06_PO. The right pane shows the output of a DRC analysis command. The output lists 64 sub-areas, all of which are complete with 0 violations.

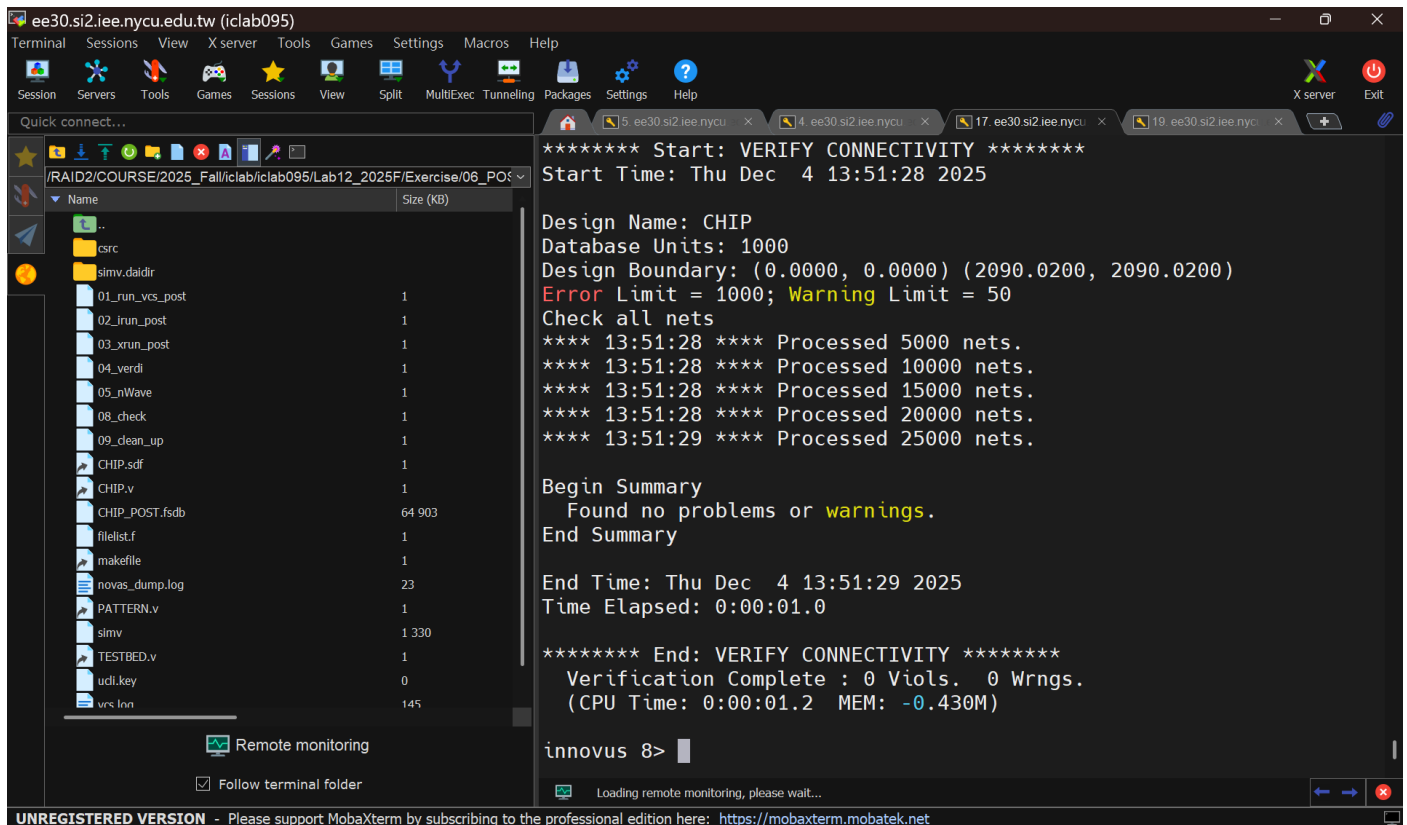
```
VERIFY DRC ..... Sub-Area: {1589.760 1589.760 1854.720 1854.720}
VERIFY DRC ..... Sub-Area : 55 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1854.720 1589.760 2090.020 1854.720}
VERIFY DRC ..... Sub-Area : 56 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {0.000 1854.720 264.960 2090.020} 57
VERIFY DRC ..... Sub-Area : 57 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {264.960 1854.720 529.920 2090.020} 5
VERIFY DRC ..... Sub-Area : 58 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {529.920 1854.720 794.880 2090.020} 5
VERIFY DRC ..... Sub-Area : 59 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {794.880 1854.720 1059.840 2090.020}
VERIFY DRC ..... Sub-Area : 60 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1059.840 1854.720 1324.800 2090.020}
VERIFY DRC ..... Sub-Area : 61 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1324.800 1854.720 1589.760 2090.020}
VERIFY DRC ..... Sub-Area : 62 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1589.760 1854.720 1854.720 2090.020}
VERIFY DRC ..... Sub-Area : 63 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1854.720 1854.720 2090.020 2090.020}
VERIFY DRC ..... Sub-Area : 64 complete 0 Viols.

Verification Complete : 0 Viols.

*** End Verify DRC (CPU: 0:00:04.7 ELAPSED TIME: 5.00 MEM: 4.0M)
innovus 8>
```

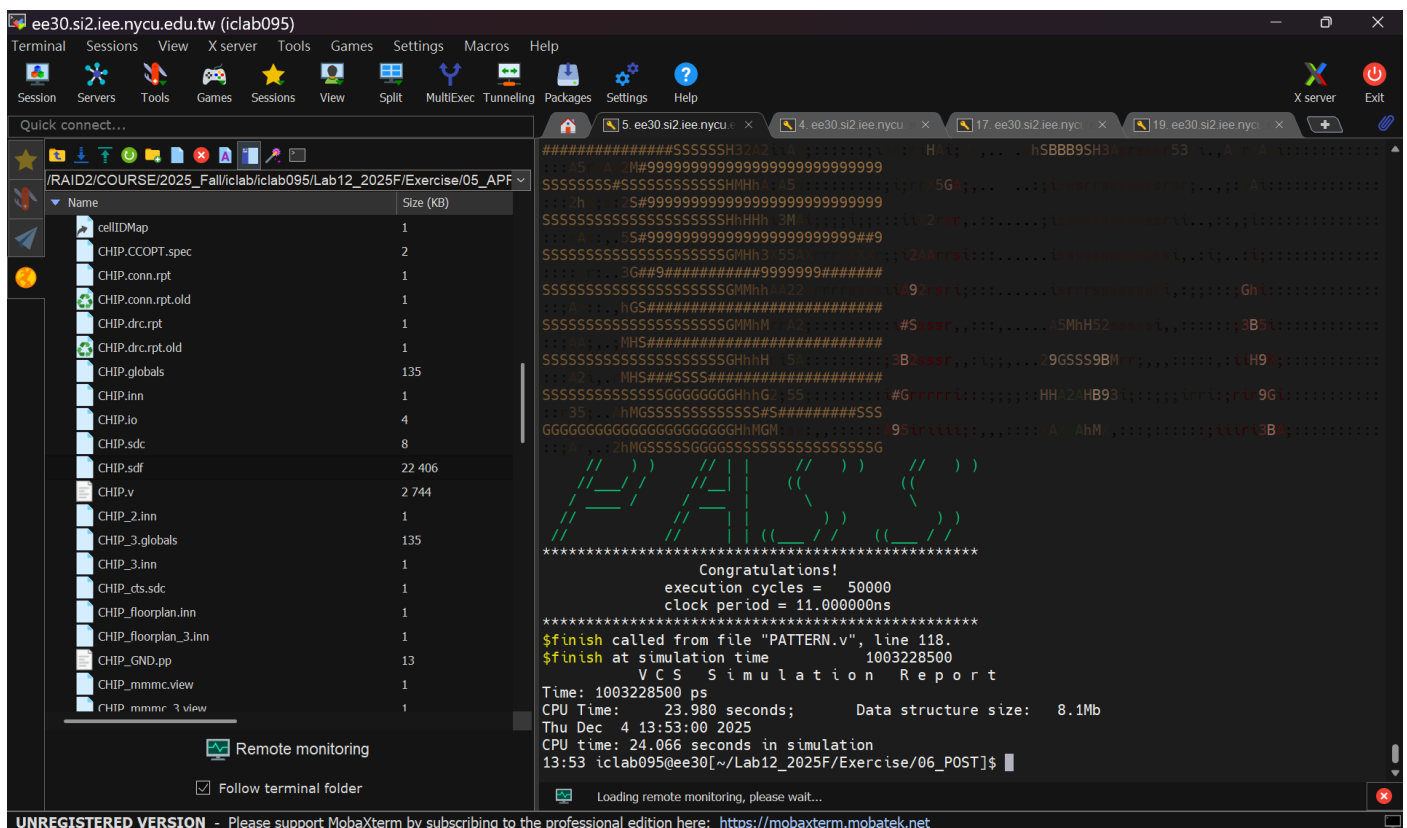
7. LVS result

Instructions: After completing the final layout, run the analysis and capture a screenshot of the results displayed in the terminal (including the number of violations).



8. Post Layout simulation result

Instructions: After completing the final layout, run the post-simulation in 06_POST and capture a screenshot of the results displayed in the terminal



9. Power result

After running the analysis, capture a screenshot of the results in the terminal (including Total Power).

```
ee30.si2.tee.nycu.edu.tw (iclab095)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help
Quick connect...
/RAID2/COURSE/2025_Fall/iclab/iclab095/Lab12_2025F/Exercise/05_APF
Name Size (KB)
..
technonly.d
.top.power.db 4
cellIDMap 762
CHIP.rpt 2 174
power.db 10 073
power.db.cnstr.td 1
static_default_ground_rail.ptiavg 4
static_default_power_rail.ptiavg 4
static_GND.ptiavg 678
static_VCC.ptiavg 678
technonly.report 146
technonly.summary 1 433
voltus_power.ptifiles 1

Remote monitoring
Follow terminal folder

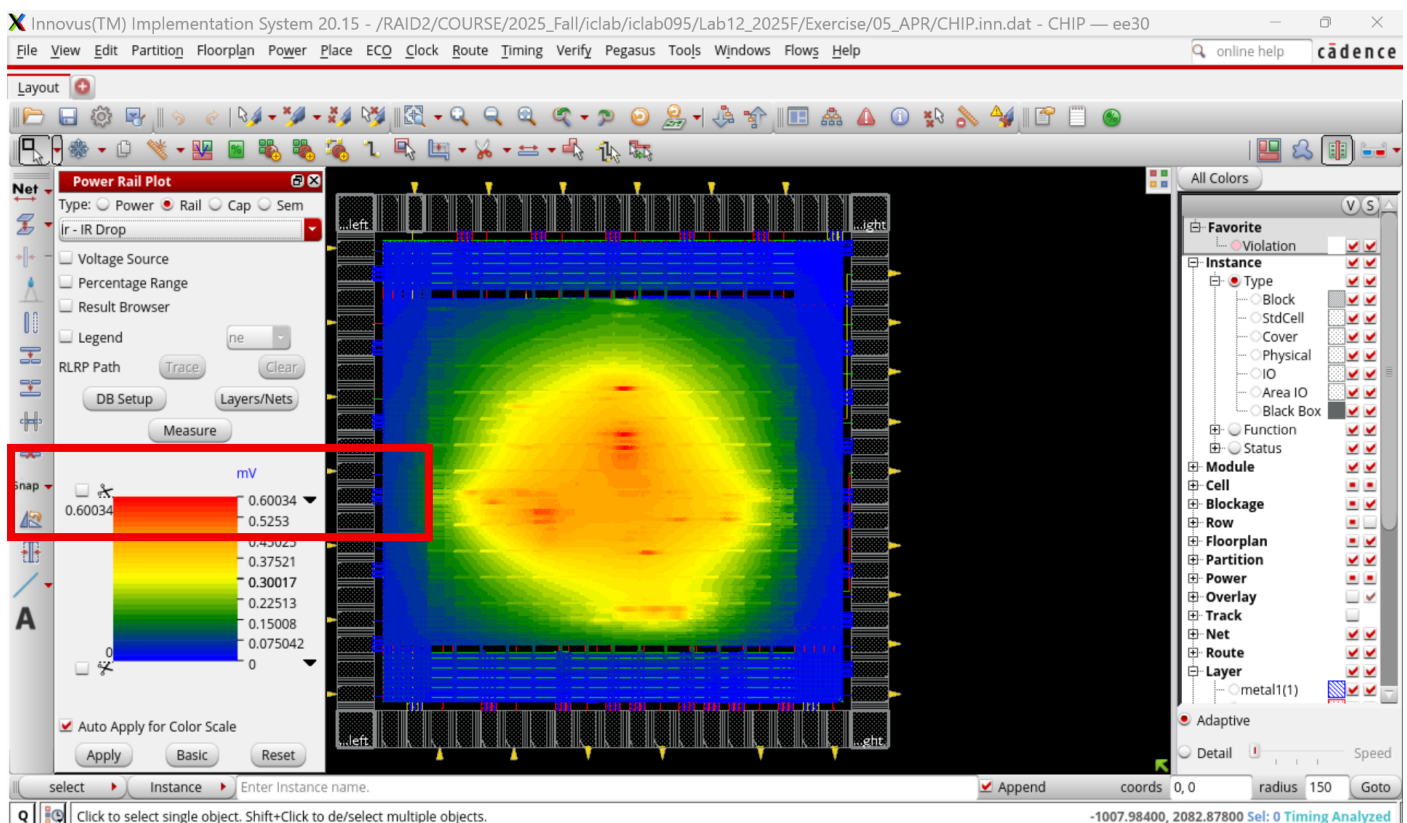
UNREGISTERED VERSION - Please support MobaXterm by subscribing to the professional edition here: https://mobaxterm.mobatek.net

Ended Power Computation: (cpu=0:00:01, real=0:00:01, mem(process/total/peak)=2458.40MB/5066.45MB/2476.97MB)
Begin Processing User Attributes
Ended Processing User Attributes: (cpu=0:00:00, real=0:00:00, mem(process/total/peak)=2458.40MB/5066.45MB/2476.97MB)
Ended Power Analysis: (cpu=0:01:24, real=0:20:47, mem(process/total/peak)=2458.40MB/5066.45MB/2476.97MB)
Begin Boundary Leakage Calculation
Ended Boundary Leakage Calculation: (cpu=0:00:00, real=0:00:00, mem(process/total/peak)=2458.40MB/5066.45MB/2476.97MB)
Begin Static Power Report Generation
48 instances have no static power
*
Total Power
-----
Total Internal Power: 3.80290034 50.6892%
Total Switching Power: 3.67629462 49.0016%
Total Leakage Power: 0.02319526 0.3092%
Total Power: 7.50239022
-----
Ended Static Power Report Generation: (cpu=0:00:00, real=0:00:00, mem(process/total/peak)=2458.41MB/5066.45MB/2476.97MB)
Begin Creating Binary Database
Ended Creating Binary Database: (cpu=0:00:00, real=0:00:01, mem(process/total/peak)=2973.19MB/5836.87MB/2973.19MB)
Output file is power_log/CHIP.rpt
Loading remote monitoring, please wait...
```

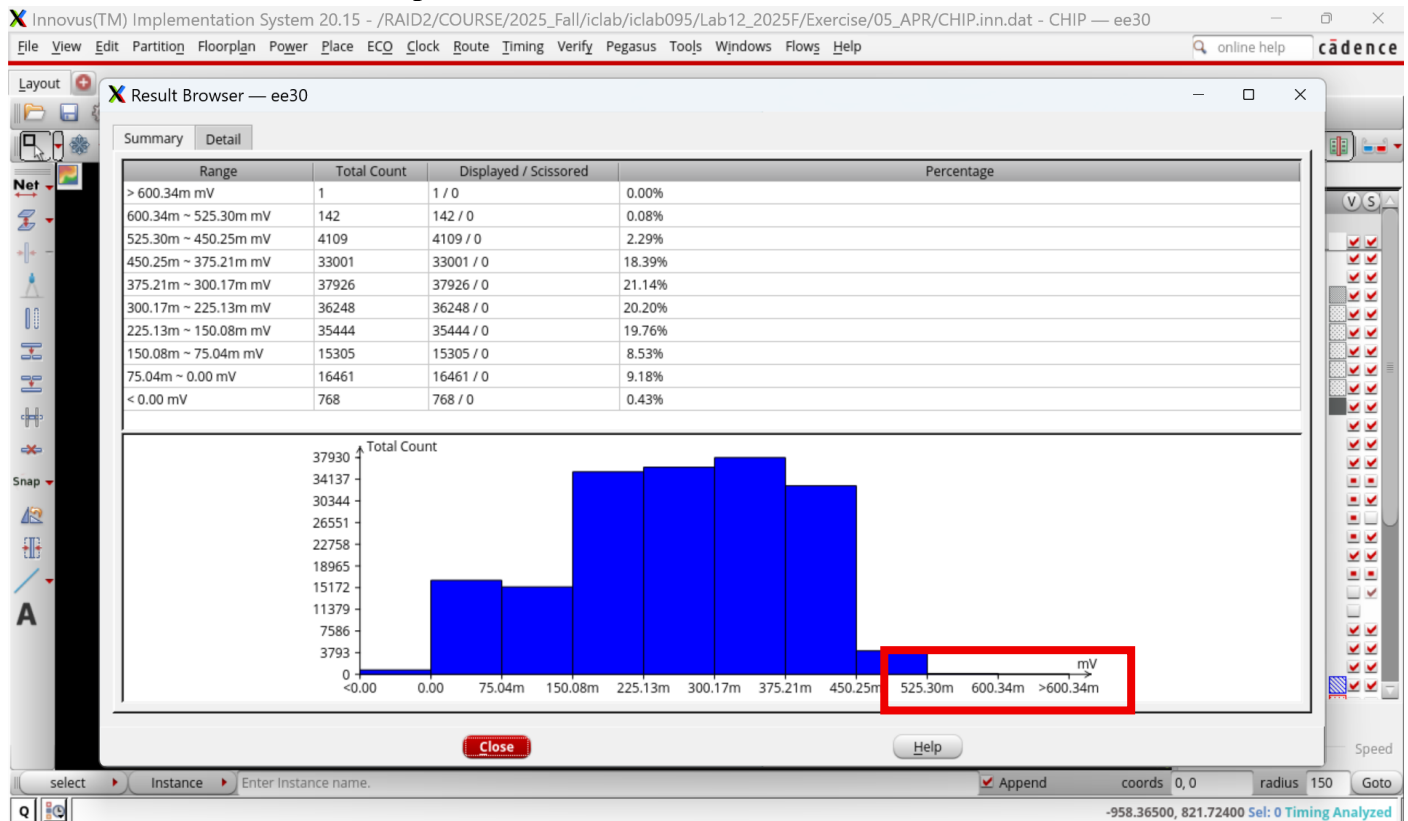
10. IR Drop Results :

Instructions: After completing the analysis, capture a screenshot of the results in Innovus. **Highlight the maximum value of the IR drop and provide a brief explanation of the method you used to mitigate the IR drop issue.**

maximum value of the IR drop: 0.60034mV



maximum value of the IR drop: 0.60034mV



Explanation:

目前的作法是依照助教的建議，增加 Core PowerPad 與 IO PowerPad 的數量。其中，IO PowerPad 共配置 8 組，而 Core PowerPad 則配置 16 組。

在 PowerPad 的擺放上，我也盡量依助教所述，避免將 PowerPad 集中於晶片的單一邊，而是盡可能均勻分佈於四周，以避免造成某些區域的邏輯閘因電源距離過遠而產生不利影響。

此外，目前的 power stripe 寬度設定為 4。本來有考慮將 stripe 與 stripe 之間間距（原為 100）再縮小，但由於目前的 IR Drop 檢查已通過，因此暫時尚未調整。

若後續需要再優化，則會考慮進一步增加 PowerPad 數量，或縮小 stripe 間距，不過此部分仍需視 DRC 是否能通過而定。